

Every positive multiple of four has a Liouville-negative representation

For a positive integer n , let $\Omega(n)$ be the number of prime factors of n , counted with multiplicity, and put

$$\lambda(n) = (-1)^{\Omega(n)} \in \mathbb{Z}.$$

In particular, $\Omega(1) = 0$. Write $\text{HasRepresentation}(N)$ for the assertion that there exist $a, b \in \mathbb{N}$ such that

$$0 < a, \quad 0 < b, \quad N = a + b, \quad \lambda(a) = \lambda(b) = -1.$$

Equal witnesses are allowed.

Theorem. For every $m \in \mathbb{N}$ with $m > 0$,

$$\text{HasRepresentation}(4m).$$

1. Elementary Liouville facts

Positive prime factorization gives, for all positive integers u, v ,

$$\Omega(uv) = \Omega(u) + \Omega(v), \quad \lambda(uv) = \lambda(u)\lambda(v). \quad (1)$$

No coprimality assumption is needed: multiplicities add even when the two factors have common prime divisors. Also,

$$\lambda(1) = 1, \quad \lambda(n) \in \{1, -1\} \quad (n > 0), \quad \lambda(r) = -1 \quad (r \text{ prime}). \quad (2)$$

Consequently,

$$\lambda(2) = \lambda(3) = \lambda(5) = -1, \quad \lambda(4) = 1, \quad \lambda(t^2) = 1 \quad (t > 0). \quad (3)$$

These are the positive-argument factorization identities; no value of $\lambda(0)$ will be used.

We also use two elementary consequences of prime factorization. A prime dividing a product divides at least one factor, so cancellation by a nonzero residue is valid modulo a prime. Every integer greater than 1 has a prime divisor: its least divisor greater than 1 must be prime, since a nontrivial proper factor of that divisor would be a smaller divisor greater than 1.

2. Reduction to odd primes

It suffices to prove

$$\text{HasRepresentation}(4p) \quad \text{for every odd prime } p. \quad (4)$$

Indeed, suppose (4) holds and let $m > 0$.

If $\lambda(m) = 1$, choose

$$a = b = 2m.$$

Both witnesses are positive, their sum is $4m$, and (1) gives $\lambda(2m) = -1$. This includes $m = 1$, with witnesses 2, 2.

Next suppose $\lambda(m) = -1$ and m is even. Write $m = 2t$; the division is exact and $t > 0$. The following two sign cases give positive witnesses of total $8t = 4m$:

condition	a	b
$\lambda(t) = 1$	$3t$	$5t$
$\lambda(t) = -1$	$4t$	$4t$

In the first row, (1) and the prime values at 3, 5 give both signs -1 . In the second, $\lambda(4) = 1$ gives both signs -1 . Thus the displayed construction works for every positive even m ; under the present additional assumption $\lambda(m) = -1$, only the first row is needed.

It remains to consider odd m with $\lambda(m) = -1$. Since $\lambda(1) = 1$, we have $m > 1$. Choose a prime divisor p of m , and write

$$m = pd, \quad d > 0.$$

The prime p is odd, since it divides the odd integer m . By (1),

$$-1 = \lambda(m) = \lambda(p)\lambda(d) = -\lambda(d),$$

so $\lambda(d) = 1$. If $a, b > 0$ witness (4), then $da, db > 0$,

$$da + db = d(4p) = 4m, \quad \lambda(da) = \lambda(d)\lambda(a) = -1, \quad \lambda(db) = -1.$$

There is no requirement that d be coprime to p , a , or b . These cases prove the reduction. Conversely, the theorem for every positive m immediately includes every odd prime $m = p$.

We now prove (4).

3. Consequences of a missing representation

For the next two sections, let p be an odd prime with $p \neq 3$, and let

$$f: \mathbb{Z}_{>0} \longrightarrow \{1, -1\}$$

be completely multiplicative, with

$$f(2) = f(p) = -1.$$

Assume, for contradiction, that there are no positive integers a, b with

$$a + b = 4p, \quad f(a) = f(b) = -1. \quad (5)$$

All arguments of f below are positive integers.

Complete multiplicativity and the nonzero sign values imply

$$f(1) = 1, \quad f(t^2) = 1 \quad (t > 0), \quad f(4) = 1.$$

Since $p + 3p = 4p$ and $f(p) = -1$, assumption (5) forces $f(3p) = 1$. Therefore

$$f(3) = -1. \quad (6)$$

More generally, whenever $0 < a < 4p$ and $f(a) = -1$, the positive complement $4p - a$ has sign 1.

Two useful implications follow by scaling positive pairs:

$$0 < u < p, \quad f(u) = -1 \implies f(p - u) = 1; \quad (A)$$

otherwise $4u$ and $4(p - u)$ would be positive negative-sign witnesses of total $4p$, since $f(4) = 1$. Likewise,

$$0 < v < 2p, \quad f(v) = 1 \implies f(2p - v) = -1; \quad (C)$$

otherwise $2v$ and $2(2p - v)$ would both have sign -1 , since $f(2) = -1$.

Combining (A) with (C) applied to $v = p - u$ gives

$$0 < u < p, \quad f(u) = -1 \implies f(p + u) = -1. \quad (B)$$

Here $0 < p - u < p < 2p$, so this application of (C) is within its stated domain.

For completeness, the same implications give local ternary identities. If $0 < 3x < p$, then

$$f(p + 3x) = -f(x) = f(3x), \quad f(p - x) = -f(x). \quad (7)$$

To see the first identity, if $f(x) = -1$, then (A) gives $f(p - x) = 1$. Equation (6) makes $3(p - x)$ negative in sign, so its complement $p + 3x$ has sign $1 = -f(x)$. If instead $f(x) = 1$, then $f(3x) = -1$, and (B) applied to $3x < p$ gives $f(p + 3x) = -1 = -f(x)$.

For the second identity in (7), the case $f(x) = -1$ is (A). If $f(x) = 1$ and also $f(p - x) = 1$, then both $p + 3x$ and $3(p - x)$ have sign -1 , contradicting (5). Every complement just used is positive: $p - x > 0$, $0 < 3(p - x) < 3p < 4p$, and $0 < p + 3x < 2p$.

The local identity (7) is not being assumed outside its interval. The following descent proves the full identity separately.

4. Ternary gap descent and full antireflection

Call a pair (x, y) a defect if

$$x > 0, \quad y > 0, \quad x + y = p, \quad f(x) = f(y) = 1.$$

In particular, both x and y lie strictly below p . We prove that no defect exists.

4.1. Neither entry of a defect is divisible by three

Suppose $x = 3u$ in a defect. Then $0 < u < p$, and (6) gives $f(u) = -1$. By (A), $f(p - u) = 1$, whence

$$f(3p - x) = f(3(p - u)) = -1.$$

On the other hand, (C) applied to $v = y$, where $0 < y < p < 2p$, gives

$$f(p + x) = f(2p - y) = -1.$$

These arguments are positive and have total $4p$. More precisely,

$$2p < 3p - x < 3p, \quad p < p + x < 2p.$$

This contradicts (5). Thus $3 \nmid x$; the same argument with x, y exchanged gives $3 \nmid y$.

4.2. Exact division gives another defect

Because p is prime and $p \neq 3$, we also have $3 \nmid p$. The nonzero residues of x, y modulo 3 must therefore agree: distinct nonzero residues would be 1, 2, whose sum is 0 modulo 3. It follows that

$$p + x = 2x + y \equiv 0 \pmod{3}, \quad p + y = x + 2y \equiv 0 \pmod{3}.$$

Thus the divisions in

$$x' = \frac{p+x}{3}, \quad y' = \frac{p+y}{3} \quad (8)$$

are exact. Both new integers are positive, and

$$x' + y' = \frac{2p + x + y}{3} = p.$$

By (C), applied to y and x respectively,

$$f(p+x) = f(p+y) = -1.$$

Since $p+x = 3x'$, $p+y = 3y'$, and $f(3) = -1$, complete multiplicativity gives

$$f(x') = f(y') = 1.$$

Hence (x', y') is again a defect. Its entries, being positive and summing to p , are both strictly less than p .

4.3. The positive integer gap decreases

As p is odd, a defect cannot have $x = y$. If any defect exists, exchange its entries to arrange $x < y$, and choose one for which the positive integer gap

$$\delta = y - x$$

is minimal. Such a minimum exists, since the possible pairs form a finite set. Construction (8) preserves the order and gives

$$0 < y' - x' = \frac{y-x}{3} = \frac{\delta}{3} < \delta.$$

The new gap is an integer because both divisions in (8) have already been proved exact. We have obtained another defect with a smaller positive integer gap, a contradiction.

There are therefore no positive-positive sign pairs of total p . There are also no negative-negative sign pairs of that total, by (A). Since the only signs are 1, -1 , we conclude that

$$f(p-n) = -f(n) \quad \text{for every } 0 < n < p. \quad (9)$$

5. Antireflection forces multiplicativity on nonzero residues

We now prove an independent lemma: if p is an odd prime, f is completely multiplicative on positive integers with values in $\{1, -1\}$, and (9) holds, then its values on $1, \dots, p-1$ define a multiplicative function on nonzero residues modulo p . In particular that function is positive on every nonzero square.

This lemma requires no assumption about $f(p)$.

Let

$$R_p = (\mathbb{Z}/p\mathbb{Z}) \setminus \{0\},$$

and write n for the residue of an integer n . For $z \in R_p$, let $n \in \{1, \dots, p-1\}$ be its unique representative and define

$$F(z) = f(n).$$

This definition does not assert that f is periodic. Prime cancellation ensures that products of nonzero residues are nonzero. Also, multiplication by a nonzero residue is injective on the finite set R_p , hence is a permutation. It therefore attains 1, so each nonzero residue has an inverse.

By (9),

$$F(-z) = -F(z) \quad (z \in R_p), \quad F(1) = 1, \quad F(-1) = -1. \quad (10)$$

Call $a \in R_p$ a good multiplier if

$$F(az) = F(a)F(z) \quad \text{for every } z \in R_p. \quad (11)$$

The residues 1 and -1 are good. The product of two good multipliers a, b is good: goodness first gives $F(ab) = F(a)F(b)$, and then, for every $z \in R_p$,

$$F(abz) = F(a)F(bz) = F(a)F(b)F(z) = F(ab)F(z). \quad (12)$$

5.1. A cyclic short-multiple lemma

Let $2 \leq n < p$, and let $z \in R_p$. We claim that there are integers k, d satisfying

$$0 < |k| < n, \quad d > 0, \quad nd < p, \quad kz = \bar{d}. \quad (13)$$

Consider the n residues

$$0, z, 2z, \dots, (n-1)z.$$

They are distinct. Indeed, an equality between the multiples with distinct indices would, by cancellation of z , make p divide a nonzero integer of absolute value less than $n < p$.

Sort their representatives in $\{0, \dots, p-1\}$ as

$$c_1 < c_2 < \dots < c_n.$$

The circular gaps are

$$c_2 - c_1, c_3 - c_2, \dots, c_n - c_{n-1}, p + c_1 - c_n.$$

All n gaps are positive integers, and their sum is p . Thus some gap d satisfies $d \leq p/n$. Since p is prime and $2 \leq n < p$, the integer n does not divide p . The integer d therefore cannot equal p/n , so

$$1 \leq d < \frac{p}{n}, \quad nd < p. \quad (14)$$

For each representative c_j , retain its original index $i_j \in \{0, \dots, n-1\}$, so that c_j represents $i_j z$. If the chosen gap is $c_{j+1} - c_j$, take $k = i_{j+1} - i_j$. If it is the wrapping gap $p + c_1 - c_n$, take $k = i_1 - i_n$. In either case the indices are distinct, so $0 < |k| < n$. In the first case their representatives differ by d ; in the wrapping case they differ by $d - p$. Both differences are congruent to d modulo p . Thus $\bar{k}z = \bar{d}$, proving (13), including the wrapping case.

5.2. The least bad multiplier cannot exist

Suppose some multiplier is not good, and choose the least representative

$$n \in \{1, \dots, p-1\}$$

of a bad multiplier. Since 1 is good, $2 \leq n < p$. Every positive integer $j < n$ represents a good multiplier. Because -1 is good and good multipliers are closed under products, every nonzero integer k with $|k| < n$ also represents a good multiplier. Here negative integers are used only as representatives of residue classes, never as arguments of f .

Fix any $z \in R_p$, and choose k, d from (13). Since d and nd are positive integers less than p , ordinary complete multiplicativity gives

$$F(\overline{nd}) = f(nd) = f(n)f(d) = F(\bar{n})F(\bar{d}). \quad (15)$$

No product has been reduced past p in this application of f .

On the other hand, $\bar{k}$ is good and $\bar{k}z = \bar{d}$, so

$$F(\bar{d}) = F(\bar{k})F(z). \quad (16)$$

Also $nz \neq 0$, and $\bar{k}nz = \overline{nd}$. Goodness of $\bar{k}$ therefore gives

$$F(\overline{nd}) = F(\bar{k})F(nz). \quad (17)$$

Combining (15)–(17) and cancelling the nonzero sign $F(\bar{k})$ yields

$$F(\bar{n}z) = F(\bar{n})F(z).$$

The residue z was arbitrary, so $\bar{n}$ is good, contradicting its choice. All multipliers are therefore good, and

$$F(ab) = F(a)F(b) \quad (a, b \in R_p). \quad (18)$$

In particular,

$$F(z^2) = F(z)^2 = 1 \quad (z \in R_p). \quad (19)$$

The strict bound $nd < p$ from the circular-gap argument is what justifies the passage from ordinary multiplicativity in (15) to the residue identity (18).

5.3. The obstruction supplied by a nonzero square

Combining §§3–5, under the missing-pair assumption (5), the induced function satisfies both

$$F(-1) = -1 \quad \text{and} \quad F(z^2) = 1 \quad (z \in R_p). \quad (20)$$

Consequently, if a positive integer $r < p$ has $f(r) = -1$ and is a nonzero square modulo p , then

$$-1 = f(r) = F(\bar{r}) = 1,$$

a contradiction. Only the values of f at the representatives $1, \dots, p-1$ define F ; no periodicity of f at larger integers has been used.

6. A small quadratic-residue prime for $p \equiv 3 \pmod{4}$

Lemma. If $p \geq 7$ is prime and $p \equiv 3 \pmod{4}$, there exists a prime r such that

$$r \leq \frac{p+1}{4} < p$$

and r is a nonzero square modulo p .

Proof. The integer

$$t = \frac{p+1}{4}$$

is at least 2. Choose a prime divisor r of t , and write $t = rs$ with $s \geq 1$. All these divisions are exact, and

$$r \geq 2, \quad r \leq t < p, \quad p = 4rs - 1. \quad (21)$$

We will prove directly that this r is a square modulo p .

Set

$$h = \frac{p-1}{2} = 2rs - 1.$$

For each $j \in \{1, \dots, h\}$, let α_j be the unique representative of rj modulo p in

$$\{-h, \dots, -1, 1, \dots, h\}.$$

It exists because $p \nmid rj$: both r and j are positive and less than p , and p is prime.

The absolute values $|\alpha_j|$ permute $1, \dots, h$. Indeed, if $|\alpha_i| = |\alpha_j|$, then $ri \equiv rj$ or $ri \equiv -rj \pmod{p}$.

Cancellation of r gives $i \equiv j$ or $i + j \equiv 0 \pmod{p}$. The first implies $i = j$, since both indices lie between 1 and $h < p$. The second is impossible, since

$$2 \leq i + j \leq 2h = p - 1.$$

Thus the absolute values are distinct, and there are h of them in a set of size h .

Let E be the number of negative α_j . Taking products gives

$$r^h h! \equiv \prod_{j=1}^h \alpha_j = (-1)^E h! \pmod{p}.$$

Every factor of $h!$ lies strictly between 0 and p , so $h!$ is nonzero modulo p and can be cancelled. Therefore

$$r^h \equiv (-1)^E \pmod{p}. \quad (22)$$

We now determine the parity of E by a finite floor calculation. The representative α_j is negative precisely when the fractional part of rj/p exceeds $1/2$. The fractional part is not 0, since $p \nmid rj$, and is not $1/2$, since a residue divided by the odd integer p cannot equal $1/2$. Hence

$$E = \sum_{j=1}^h \left(\left\lfloor \frac{2rj}{p} \right\rfloor - 2 \left\lfloor \frac{rj}{p} \right\rfloor \right). \quad (23)$$

Each summand is 0 or 1, and is 1 exactly for a negative representative. In particular, E has the same parity as

$$S = \sum_{j=1}^h \left\lfloor \frac{2rj}{p} \right\rfloor.$$

Since

$$0 < \frac{2rj}{p} \leq \frac{r(p-1)}{p} < r,$$

we may count the integer levels $k = 1, \dots, r-1$ to obtain

$$S = \sum_{k=1}^{r-1} \# \left\{ j \in \{1, \dots, h\} : \frac{2rj}{p} \geq k \right\}. \quad (24)$$

For each such k , (21) gives

$$\frac{kp}{2r} = 2ks - \frac{k}{2r}.$$

Here $0 < k/(2r) < 1$, so the threshold is nonintegral and

$$\left\lfloor \frac{kp}{2r} \right\rfloor = 2ks - 1. \quad (25)$$

Moreover,

$$1 \leq 2ks - 1 \leq 2s(r-1) - 1 = h - 2s < h.$$

Thus there is no truncation at either endpoint of $\{1, \dots, h\}$. Because the threshold is nonintegral, the number of indices in (24) is exactly

$$h - \left\lfloor \frac{kp}{2r} \right\rfloor = (2rs - 1) - (2ks - 1) = 2s(r - k). \quad (26)$$

Every term in (26) is even. Consequently S , and hence E , is even. Equation (22) gives

$$r^h \equiv 1 \pmod{p}.$$

Finally, $2t = h + 1$, so the explicit integer $A = r^t$ satisfies

$$A^2 = r^{2t} = r^{h+1} \equiv r \pmod{p}.$$

Since $r \not\equiv 0 \pmod{p}$, this is a nonzero square, proving the lemma. $\square$

7. Representations for every odd prime

We apply the preceding arguments with $f = \lambda$. Equations (1)–(3) give complete multiplicativity, the two sign values, and $f(2) = f(p) = -1$ whenever p is prime.

7.1. Primes congruent to three modulo four

Let $p \geq 7$ be prime with $p \equiv 3 \pmod{4}$. Suppose $\text{HasRepresentation}(4p)$ fails. All hypotheses of §§3–5 hold, so a negative-sign nonzero square strictly below p would contradict §5.3.

The lemma in §6 supplies precisely such an integer: it gives a prime $r < p$ that is a nonzero square modulo p , while $\lambda(r) = -1$. This contradiction proves

$$\text{HasRepresentation}(4p).$$

7.2. Primes congruent to one modulo four

Let $p \equiv 1 \pmod{4}$ be prime. Then $p \geq 5$, so p is odd and $p \neq 3$. If $\text{HasRepresentation}(4p)$ failed, §§3–5 would again yield (20). We show directly that -1 is a nonzero square modulo p , contradicting (20).

Every residue in R_p has an inverse, as proved in §5. Pair each residue with its inverse. The self-inverse residues satisfy $a^2 = 1$, hence

$$(a - 1)(a + 1) = 0.$$

Prime cancellation shows that they are exactly 1 and -1 , which are distinct because p is odd. Every other inverse pair has product 1. Therefore the product of all nonzero residues is -1 .

Put $h = (p - 1)/2$. Pairing the integer representatives instead as $j, p - j$, for $j = 1, \dots, h$, gives

$$\prod_{a=1}^{p-1} a = \prod_{j=1}^h j \overline{p-j} = (-1)^h (\overline{h!})^2.$$

Since $p \equiv 1 \pmod{4}$, the integer h is even. Comparing the two products yields

$$(\overline{h!})^2 = -1.$$

The residue $\overline{h!}$ is nonzero because all its factors lie in $\{1, \dots, p - 1\}$. Thus (19) gives $F(-1) = 1$, whereas (10) gives $F(-1) = -1$, a contradiction. This proves $\text{HasRepresentation}(4p)$ in this case also.

7.3. The prime three

For $p = 3$, take the positive witnesses

$$12 = 5 + 7.$$

The integers 5, 7 are prime: a composite among them would have a proper factor between 2 and its square root, hence an integer factor 2, but both are odd. Thus $\lambda(5) = \lambda(7) = -1$.

Every odd prime is either 3, congruent to 1 modulo 4, or congruent to 3 modulo 4 and at least 7. The three cases prove (4).

8. Conclusion for every positive multiplier

The odd-prime assertion (4), together with the reduction in §2, proves the theorem for every $m > 0$. Explicitly, the diagonal witnesses $2m, 2m$ handle $\lambda(m) = 1$; the positive scaled pairs from $8 = 3 + 5 = 4 + 4$ handle even m ; and for odd m with $\lambda(m) = -1$, a prime factorization $m = pd$ with $\lambda(d) = 1$ scales a representation of $4p$ to one of $4m$.

In every case the witnesses are positive natural numbers, their sum is exactly $4m$, and each has Liouville value -1 . No distinctness, parity, or coprimality condition is imposed on the witnesses. $\square$

Mathematical sources

- The definitions are those of `Statement/Definitions.lean`, `omega` and `lambda`, and `Statement/Partial.lean`, `HasSignedRepresentation` and `HasRepresentation`. The elementary identities in §1 are `omega_mul`, `lambda_one`, `lambda_sign`, `lambda_mul`, and `lambda_prime` in `Partial.lean`; the factor-list inputs are `Nat.perm_primeFactorsList_mul` and `Nat.primeFactorsList_prime`.
- The local arguments are from `nl/descent/proof-attempt-v1.md`, with the following scope: its §0 baseline supplies §1 above; its §§1–2 supply §3; its §3 supplies §4; its §§4–5 supply §5; its §6 supplies §6; and its §§7–8 supply the prime cases, reduction, and final assembly in §§2, 7–8. The floor-parity and inverse-pairing arguments are proved here directly; neither quadratic reciprocity nor a sum-of-two-squares theorem is invoked.